

End-to-End Process · Order-to-Cash (O2C)

End-to-End Process Specification

Step-level specification: People · Process · Data / CRUD · Technology · PCF 3.5 + 4.4 + 9.4

Step · PCF	Pattern	M3 H%	People	Process	Data · CRUD	Technology
PCF 3.5 · Manage Sales Orders — Order Capture						
Accept & Validate Sales Order 3.5.4	Decision	80%	Sales / Order Management — Commercial authority AI validates against customer master, credit limit, and availability. Human reviews exceptions and approves non-standard orders.	Receive sales order. Validate customer, credit, and availability. Accept or reject. Flag non-standard terms for human review.	C Sales Order · ERP — Order Mgmt R Customer Master · ERP — CRM R Credit Limit · ERP — Finance R Product Master · ERP — MM	ERP Order Mgmt — SAP SD / Oracle OM CRM — Salesforce / SAP CRM AI Validation — Credit and availability check
Determine Availability 3.5.4	Transaction	35%	AI — System automated Real-time ATP check against inventory and production schedule. Human intervenes only on constraint exceptions.	Check inventory availability. Run ATP logic. Confirm availability date. Flag shortfalls to backorder processing.	R Inventory Master · ERP — MM / WM U Sales Order · ERP — Order Mgmt	ERP ATP — SAP SD / Oracle ATP AI Scheduling — Demand vs supply matching
Determine Fulfillment Process 3.5.4	Decision	80%	Sales Operations — Fulfillment authority AI proposes fulfillment route. Human approves routing for complex or high-value orders.	Determine delivery method, shipping route, warehouse source. Assign fulfillment instruction to Materials. Set delivery commitment date.	U Sales Order · ERP — Order Mgmt C Fulfillment Instruction · ERP — WM R Delivery Capabilities · ERP — Logistics	ERP Logistics — SAP LE / Oracle Logistics TMS — Transportation Management AI Routing — Fulfillment optimization
Confirm Sales Order 3.5.4	Document	40%	AI — System automated Generates and distributes order confirmation. Human reviews high-value or custom orders.	Generate order confirmation. Apply agreed pricing and delivery terms. Distribute to customer via EDI or portal.	C Order Confirmation · ERP — Order Mgmt R Sales Order · ERP — Order Mgmt R Pricing Rules · ERP — Pricing	ERP Order Mgmt — SAP SD / Oracle OM EDI / Customer Portal — Confirmation distribution
Process Backorders & Updates 3.5.4	Exception	82%	Sales / Customer Service AI flags backorder. Human determines resolution — partial shipment, substitute, or cancellation.	Identify stock shortfall. Determine resolution. Notify customer of delay and options. Update order with revised commitment or cancellation.	U Sales Order · ERP — Order Mgmt R Inventory Master · ERP — MM / WM	ERP Order Mgmt — Backorder management Customer Portal — Customer notification AI Communication — Draft customer update
Handle Order Inquiries 3.5.4	Exception	82%	Sales / Customer Service AI handles routine tracking inquiries. Human resolves complex issues requiring judgment.	Receive inquiry. Classify type. Provide order status or tracking. Escalate complex inquiries. Log resolution.	R Sales Order · ERP — Order Mgmt R Delivery / Tracking · TMS / Carrier	CRM — Case management Customer Portal — Self-service tracking AI Chat — Routine inquiry resolution
PCF 4.4 · Manage Logistics and Warehousing						
Pick / Pack / Ship 4.4.3	Transaction	35%	Warehouse Operations AI directs pick path and generates packing instructions. Staff execute physical pick, pack, handoff to carrier.	Receive fulfillment instruction. Pick goods. Pack per order specification. Generate shipping docs. Hand off to carrier. Post goods issue.	R Fulfillment Instruction · ERP — WM C Shipment Record · ERP — Logistics	WMS — SAP EWM / Manhattan WMS Barcode / RFID — Pick confirmation

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					U Inventory Master · ERP — MM / WM	AI Pick Path — Warehouse route optimization
Track Inventory Accuracy 4.4.3	Transaction	35%	AI — System automated Continuous inventory reconciliation post-shipment. AI flags discrepancies. Human investigates variances.	Reconcile inventory post-pick. Update stock levels. Flag discrepancies. Trigger cycle count if variance exceeds threshold.	U Inventory Master · ERP — MM / WM R Shipment Record · ERP — Logistics	WMS — Real-time inventory tracking AI Reconciliation — Variance detection
Plan & Execute Outbound Delivery 4.4.4	Transaction	35%	AI — System automated AI plans carrier selection, route, and delivery schedule. Transmits shipment to carrier. Human monitors SLA compliance.	Select carrier. Plan route. Transmit shipping instructions. Record delivery commitment. Notify customer of shipment.	C Delivery Record · TMS / ERP Logistics R Shipment Record · ERP — Logistics R Transportation Schedules · TMS	TMS — SAP TM / Oracle TMS Carrier EDI — Shipment transmission AI Route Optimization — Cost and SLA balancing
Track Carrier Delivery Performance 4.4.4	Transaction	35%	AI — System automated AI monitors carrier tracking and flags exceptions. Human intervenes on missed SLAs.	Monitor carrier tracking events. Confirm proof of delivery. Record delivery completion. Create billing trigger on confirmation.	U Delivery Record · TMS / ERP Logistics C Proof of Delivery · TMS / Carrier C Billing Trigger · ERP — Finance	TMS — Carrier tracking integration AI Monitoring — SLA exception detection
PCF 9.4 · Manage Customer Credit, Invoicing, and Accounts Receivable						
Generate Invoice 9.4.2	Document	40%	AI — System automated Auto-generated on billing trigger from delivery confirmation. AI applies pricing and contract terms. Human reviews high-value invoices.	Receive billing trigger. Generate invoice from sales order and delivery record. Apply pricing rules, discounts, and contract terms.	C Customer Invoice · ERP — AR / Billing R Sales Order · ERP — Order Mgmt R Billing Trigger · ERP — Finance R Pricing Rules · ERP — Pricing	ERP Billing — SAP FICA / Oracle AR AI Invoice Generation — Contract-aware drafting
Apply Taxes 9.4.2	Transaction	35%	AI — System automated Tax engine calculates taxes per jurisdiction, product type, and customer tax status.	Determine applicable tax jurisdiction. Calculate tax amounts. Apply to invoice. Record tax liability.	U Customer Invoice · ERP — AR / Billing R Tax Rules · Tax Engine	Tax Engine — Vertex / Avalara / SAP Tax AI Tax Classification — Jurisdiction determination
Distribute Invoice 9.4.2	Transaction	35%	AI — System automated Transmits invoice to customer via EDI, portal, or email. Records distribution and opens AR.	Transmit invoice to customer per preferred channel. Record distribution. Open AR line. Set payment due date.	U Customer Invoice · ERP — AR / Billing C AR Open Item · ERP — AR	EDI / Customer Portal — Invoice distribution ERP AR — SAP FI-AR / Oracle AR
Maintain AR 9.4.3	Transaction	35%	AI — System automated Continuous AR ledger maintenance. AI posts receipts and applies payments. Human resolves unapplied cash.	Maintain open AR items. Post incoming payments. Apply cash to oldest open items. Flag unapplied items.	U AR Open Item · ERP — AR R Customer Master · ERP — CRM	ERP AR — SAP FI-AR / Oracle AR AI Cash Application — Remittance matching
Monitor Aging 9.4.3	Transaction	35%	AI — System automated AI generates aging analysis and flags overdue accounts. Human reviews and determines collection action.	Generate AR aging report. Classify overdue accounts by risk tier. Flag accounts for collections. Notify AR team.	R AR Open Item · ERP — AR C Aging Report · ERP — AR / BI	ERP AR — Aging analysis BI / Analytics — Collections dashboard AI Risk Scoring — Payment probability

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Reconcile Accounts 9.4.3	Transaction	35%	AI — System automated AI reconciles AR ledger against bank receipts. Flags variances. Human resolves and approves period close.	Reconcile AR ledger against cash receipts and bank statements. Resolve variances. Close period. Output to Financial Reporting.	U AR Ledger · ERP — FI-GL / AR R Bank Statements · Treasury / Banking	ERP GL — SAP FI-GL / Oracle GL AI Reconciliation — Auto-match and variance detection
Manage Collections / Dunning 9.4.4	Exception	82%	Collections Manager — Recovery authority AI generates dunning notices and tracks responses. Human determines escalation — payment plan, legal referral, or write-off.	Issue dunning notices per aging tier. Track customer responses. Escalate unresponsive accounts. Negotiate payment arrangements.	C Dunning Notice · ERP — AR / Collections R AR Open Item · ERP — AR R Customer Master · ERP — CRM	Collections Platform — SAP Collections / Emagia AI Dunning — Automated notice generation AI Risk Scoring — Recovery probability
Resolve Disputes 9.4.4	Exception	82%	AR Specialist + Sales AI classifies dispute and routes to owner. Human investigates, collaborates, issues credit note or correction.	Receive dispute. Classify — pricing, quantity, quality, or duplicate. Investigate. Issue credit note or corrected invoice. Close.	U Customer Invoice · ERP — AR / Billing R Sales Order · ERP — Order Mgmt R Customer Master · ERP — CRM	Collections Platform — Dispute management CRM — Customer communication AI Classification — Dispute root cause
Process Customer Payment 9.4.5	Transaction	35%	AI — System automated Receives inbound payment. AI records, validates, and routes to cash application. Human monitors exception queue only.	Receive payment via ACH, wire, check, or card. Validate amount. Record receipt. Route to cash application.	C Payment Receipt · Treasury / Banking R AR Open Item · ERP — AR	Treasury Platform — Payment receipt Lockbox / EDI — Remittance processing AI Matching — Payment-to-invoice matching
Apply Cash 9.4.5	Transaction	35%	AI — System automated AI matches payment to open invoices and applies cash. Human resolves unapplied cash. Clears AR and confirms to customer.	Match payment remittance to open AR items. Apply cash to invoices. Post to AR ledger. Clear open items. Send payment confirmation.	U AR Open Item · ERP — AR U AR Ledger · ERP — FI-GL / AR R Payment Receipt · Treasury / Banking	ERP AR — SAP FI-AR / Oracle AR AI Cash Application — Remittance matching and posting
External Participant · Customer						
Submit Sales Order Ext	Document	40%	Customer Purchasing Submits order via EDI, web portal, email, or phone.	Prepare and submit purchase order. Specify product, quantity, delivery date, and shipping address.	C Sales Order (submitted) · Customer ERP / Portal	EDI / Customer Portal — Order submission
Receive Order Confirmation Ext	Transaction	35%	Customer Purchasing Receives and records order confirmation. Reviews delivery commitment.	Receive order confirmation. Verify quantities, pricing, and delivery date. Record in customer system.	R Order Confirmation (received) · Customer ERP / Portal	EDI / Customer Portal — Confirmation receipt
Receive Goods / Delivery Ext	Transaction	35%	Customer Receiving Receives and inspects goods. Records goods receipt. Signs proof of delivery.	Receive shipment. Inspect quantity and condition. Accept delivery. Sign proof of delivery.	C Goods Receipt (customer) · Customer ERP	Customer WMS / ERP — Goods receipt
Receive Invoice Ext	Transaction	35%	Customer AP Receives and records invoice. Routes to AP matching and payment.	Receive invoice. Validate against PO and goods receipt. Route to payment process.	R Invoice (received) · Customer AP System	Customer AP System / EDI — Invoice receipt

Step · PCF	Pattern	M3 H%	People	Process	Data · CRUD	Technology
Submit Payment Ext	Transaction	35%	Customer Finance Executes payment per agreed terms. Transmits remittance advice.	Schedule and execute payment. Transmit remittance advice. Record payment in customer system.	C Payment (transmitted) · Customer Treasury	Customer Treasury / Bank — ACH / Wire / Check
Receive Payment Confirmation Ext	Transaction	35%	Customer Finance Receives and records payment confirmation. Closes invoice in customer AP.	Receive payment confirmation. Record in customer AP. Close invoice. Reconcile against remittance.	U Invoice (closed) · Customer AP System	Customer AP System — Payment confirmation

CRUD Data Matrix · Order-to-Cash (O2C)

C = Create · R = Read · U = Update · · = No operation at this step

PCF 3.5 · Manage Sales Orders — Order Capture

Data Entity	System of Record	3.5.4 (1)	3.5.4 (2)	3.5.4 (3)	3.5.4 (4)
Sales Order	ERP — Order Mgmt	C	U	U	R
Customer Master	ERP — CRM	R	R	·	R
Credit Limit	ERP — Finance	R	·	·	·
Product Master	ERP — MM	R	R	·	·
Inventory Master	ERP — MM / WM	·	R	·	·
Fulfillment Instruction	ERP — WM	·	·	C	·
Order Confirmation	ERP — Order Mgmt	·	·	·	C
Pricing Rules	ERP — Pricing	R	·	·	R

(1) 3.5.4 Accept & Validate Sales Order · (2) 3.5.4 Determine Availability · (3) 3.5.4 Determine Fulfillment Process · (4) 3.5.4 Confirm Sales Order · Exception steps omitted

PCF 4.4 · Manage Logistics and Warehousing

Data Entity	System of Record	4.4.3 (1)	4.4.3 (2)	4.4.4 (3)	4.4.4 (4)
Fulfillment Instruction	ERP — WM	R	·	·	·
Shipment Record	ERP — Logistics	C	R	R	·
Inventory Master	ERP — MM / WM	U	U	·	·
Delivery Record	TMS / ERP Logistics	·	·	C	U
Proof of Delivery	TMS / Carrier	·	·	·	C
Billing Trigger	ERP — Finance	·	·	·	C

Data Entity	System of Record	4.4.3 (1)	4.4.3 (2)	4.4.4 (3)	4.4.4 (4)
Transportation Schedules	TMS	-	-	R	-

(1) 4.4.3 Pick / Pack / Ship · (2) 4.4.3 Track Inventory Accuracy · (3) 4.4.4 Plan & Execute Outbound Delivery · (4) 4.4.4 Track Carrier Delivery Performance

PCF 9.4 · Manage Customer Credit, Invoicing, and Accounts Receivable

Data Entity	System of Record	9.4.2 (1)	9.4.2 (2)	9.4.2 (3)	9.4.3 (4)	9.4.3 (5)	9.4.3 (6)	9.4.5 (7)	9.4.5 (8)
Customer Invoice	ERP — AR / Billing	C	U	U	R	R	-	R	-
Sales Order	ERP — Order Mgmt	R	-	-	-	-	-	-	-
Billing Trigger	ERP — Finance	R	-	-	-	-	-	-	-
Tax Rules	Tax Engine	-	R	-	-	-	-	-	-
AR Open Item	ERP — AR	-	-	C	U	R	R	R	U
Aging Report	ERP — AR / BI	-	-	-	-	C	-	-	-
AR Ledger	ERP — FI-GL / AR	-	-	-	-	-	U	-	U
Payment Receipt	Treasury / Banking	-	-	-	-	-	R	C	R
Bank Statements	Treasury / Banking	-	-	-	-	-	R	-	-
Customer Master	ERP — CRM	-	-	-	R	R	-	-	-

(1) 9.4.2 Generate Invoice · (2) 9.4.2 Apply Taxes · (3) 9.4.2 Distribute Invoice · (4) 9.4.3 Maintain AR · (5) 9.4.3 Monitor Aging · (6) 9.4.3 Reconcile Accounts · (7) 9.4.5 Process Customer Payment · (8) 9.4.5 Apply Cash · Exception steps omitted